

Electrical Machines – Transformers

Introduction

A transformer is a static electrical device used to transfer alternating electrical energy from one circuit to another through electromagnetic induction. It can increase or decrease AC voltage while maintaining the same frequency.

Working Principle

A transformer operates on Faraday's law of electromagnetic induction. When alternating current flows through the primary winding, it produces a changing magnetic flux in the core. This changing flux induces an electromotive force in the secondary winding.

Main Components

The main parts of a transformer are the magnetic core, primary winding, secondary winding, insulation system, and enclosure. Power transformers may also use oil for insulation and cooling.

Voltage Transformation

A step-up transformer increases voltage and decreases current, while a step-down transformer decreases voltage and increases current. The voltage ratio is related to the turns ratio of the primary and secondary windings.

Common Problems

Common transformer problems include overheating, insulation failure, excessive noise, low output voltage, and abnormal oil temperature in oil-filled transformers.

Safety

Before inspecting a transformer, isolate the electrical supply and follow the manufacturer's safety procedure. Do not touch energized terminals or exposed conductors.